DSA

Practical 3.2

Sentinal Search

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Batch:A2

#code

```
#include <iostream>
#include <string>
using namespace std;



struct Friend {
   string name;
   string mobile;
};



int sentinelSearch(Friend phonebook[], int n, string key) {
   Friend last = phonebook[n - 1];
   phonebook[n - 1].name = key;

   int i = 0;
   while (phonebook[i].name != key)
      i++;

   phonebook[n - 1].name = last.name;

   if (i < n - 1 || last.name == key)
      return i;
   else
      return -1;
}

// Insert a new friend while keeping the list sorted
```

```
void insertSorted(Friend phonebook[], int& n, Friend newFriend) {
    if (n >= 15) {
        cout << "Phonebook is full. Cannot add more friends.\n";
        return;
    }

    int i = n - 1;
    while (i >= 0 && phonebook[i].name > newFriend.name) {
        phonebook[i + 1] = phonebook[i];
        i--;
    }

    phonebook[i + 1] = newFriend;
    n++;
}


void displayPhonebook(Friend phonebook[], int n) {
    cout << "\nPhonebook:\n";
    for (int i = 0; i < n; i++) {
        cout << i + 1 << ". " << phonebook[i].name << " - " << phonebook[i].mobile << endl;
    }
}

int main() {
    int n;
    Friend phonebook[15]; // Max 15 friends

    cout << "Enter number of friends (max 15): ";
    cin >> n;
```

```
if (n < 0 || n > 15) {
    cout << "Invalid number. Exiting.\n";
    return 1;
}

cout << "Enter friend details (Name Mobile) in sorted order:\n";
for (int i = 0; i < n; i++) {
    cin >> phonebook[i].name >> phonebook[i].mobile;
}

int choice;
do {
    cout << "\n====== MENU ======\n";
    cout << "1. Search Friend\n";
    cout << "2. Add New Friend\n";
    cout << "3. Display Phonebook\n";
    cout << "4. Exit\n";
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice) {
        case 1: {
            string key;
            cout << "Enter friend's name to search: ";
            cin >> key;

            int index = sentinelSearch(phonebook, n, key);
            if (index != -1) {
                cout << "Friend found: " << phonebook[index].name
                     << " - " << phonebook[index].mobile << endl;
            } else {
```

```
        cout << "Friend not found.\n";
    }
    break;
}

case 2: {
    if (n >= 15) {
        cout << "Phonebook is full. Cannot add more friends.\n";
        break;
    }

    Friend newFriend;
    cout << "Enter friend's name: ";
    cin >> newFriend.name;
    cout << "Enter mobile number: ";
    cin >> newFriend.mobile;

    insertSorted(phonebook, n, newFriend);
    cout << "Friend added successfully.\n";
    break;
}

case 3:
    displayPhonebook(phonebook, n);
    break;

case 4:
    cout << "Thank You\n";
    break;

default:
```

```
            cout << "Invalid choice.\n";
        }

    } while (choice != 4);

    return 0;
}

/*
Enter number of friends (max 15): 2
Enter friend details (Name Mobile) in sorted order:
Arjun
514215
Sujal
854752

====== MENU ======
1. Search Friend
2. Add New Friend
3. Display Phonebook
4. Exit
Enter your choice: 1
Enter friend's name to search: Arjun
Friend found: Arjun - 514215

====== MENU ======
1. Search Friend
2. Add New Friend
3. Display Phonebook
4. Exit
Enter your choice: 2
```

Enter friend's name: gandhar

Enter mobile number: 54124

Friend added successfully.

====== MENU ======

1. Search Friend

2. Add New Friend

3. Display Phonebook

4. Exit

Enter your choice: 3

Phonebook:

1. Arjun - 514215

2. Sujal - 854752

3. gandhar - 54124

====== MENU ======

1. Search Friend

2. Add New Friend

3. Display Phonebook

4. Exit

Enter your choice: 4

Thank You

Process exited after 48.47 seconds with return value 0

Press any key to continue . . .

*/